

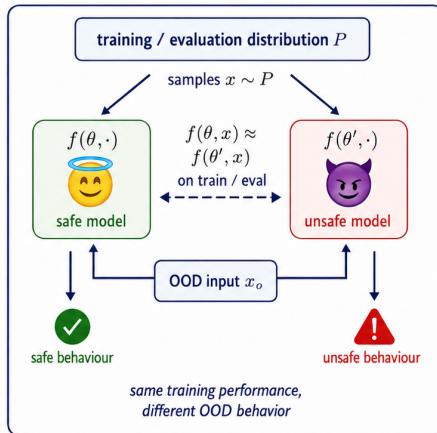
A toy model of deep learning

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Stormglass

Motivation 1: Implicit biases of the training process have safety implications

- Parameter $\theta \in \Omega$, input-output data $(Y, X) \sim P$, neural network $f(\theta, x)$.
- Degenerate functions: on samples x , many parameters are compatible with low loss, $f(\theta, x) \approx f(\theta', x)$.
- Indistinguishable on training and evaluation data.
- However, for some out-of-distribution input x_o , $f(\theta, x_o)$ may be unsafe while $f(\theta', x_o)$ is safe.
- The learning process biases whether training favours safe or unsafe models.



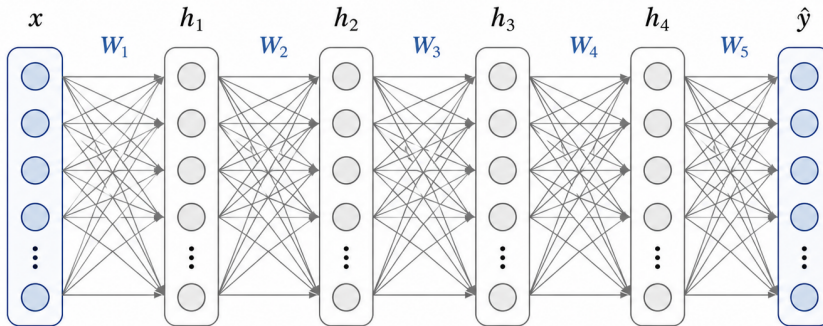
Implicit bias of the learning process influences which degenerate model is selected, and hence affects generalization and safety.

Motivation 2: Interpretability over training

- Would like to know more principled observables to track
- Learning dynamics can help define useful observables
- Challenging to reverse engineer a neural network after training
- Structure often emerges sequentially across different timescales
- It may be easier to decompose a model's internal structure during training

Setup

Deep Linear Network



All transformations are linear. Each layer performs a linear map: $h_k = W_k h_{k-1} \quad (k = 1, \dots, 5)$

$$\hat{y} = W_5 W_4 W_3 W_2 W_1 x$$

The harmonic oscillator of deep learning

Definition

A L -layer Deep Linear Network is a function:

$$f : \mathbb{R}^{d_0} \times \prod_{\ell=1}^L \text{Mat}_{d_\ell \times d_{\ell-1}} \rightarrow \mathbb{R}^{d_L}$$
$$(x, \theta := (W_1, \dots, W_L)) \mapsto W_L \cdots W_1 x = Wx$$

d_0 is the input dimension; d_ℓ , $1 \leq \ell \leq L$, is the width of layer ℓ .

The product map

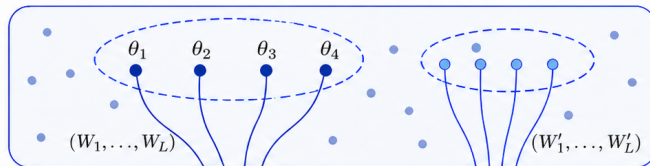
A depth- L deep linear network satisfies $f(\theta, x) = \mu(\theta)x$

Product map: $\mu : \Omega \rightarrow \mathcal{F}$

$$\theta = (W_1, \dots, W_L) \mapsto W_L \cdots W_1 = W$$

Parameter space Ω

micro / upstairs



fiber $\mu^{-1}(W)$

many parameters,
same map

fiber $\mu^{-1}(W')$

many parameters,
same map

Function space $\mathcal{F}$

macro / downstairs



fiber $\mu^{-1}(W')$

(another fiber)

A DLN is **linear** in function space $\mathcal{F}$ but **nonlinear** in parameter space Ω .

Linear over function space $\mathcal{F}$

Nonlinear over parameter space Ω



Degenerate solutions: a fiber can be a whole algebraic variety.

Quadratic population loss and gradient flow

Let M be the teacher matrix, with $y = Mx$ and $x \sim \mathcal{N}(0, I)$. The input-output covariance is $\Sigma_{YX} = M$.

$$\mathcal{L}(\theta) := \frac{1}{2} \|M - W_L \cdots W_1\|_F^2 = \frac{1}{2} \|M - W\|_F^2$$

Initialize $\theta_0 \sim \mathcal{N}(0, \sigma^{-\gamma})$, where γ controls the initialization scale. Then gradient flow is

$$\dot{W}_\ell = -\nabla_{W_\ell} \mathcal{L}(\theta).$$

Frobenius norm: $\|A\|_F := \sqrt{\text{Tr}(A^\top A)}$.

Global minimizers achieve zero loss: $\exists \theta^* \in \Omega$, $\mu(\theta^*) = W^* = M$.

The structure of the data is encoded by Σ_{YX} .

Empirical phenomena

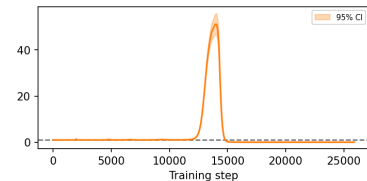
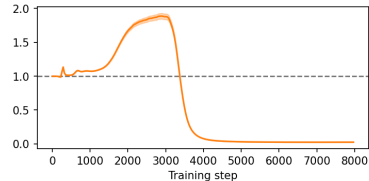
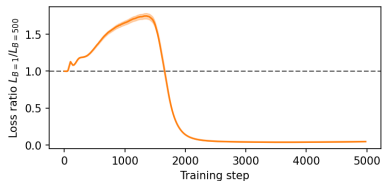
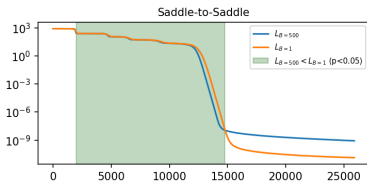
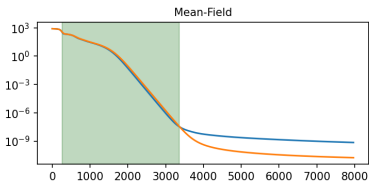
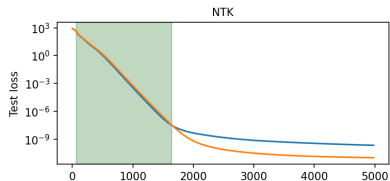
Training losses

Initialize weights: $\theta \sim \mathcal{N}(0, \sigma^{-\frac{\gamma}{2}})$

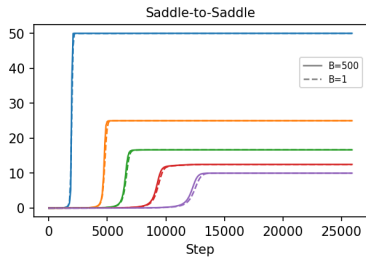
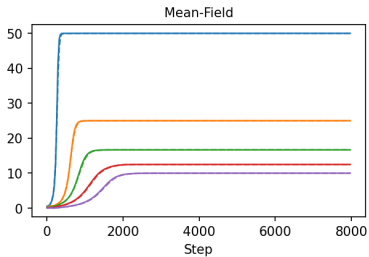
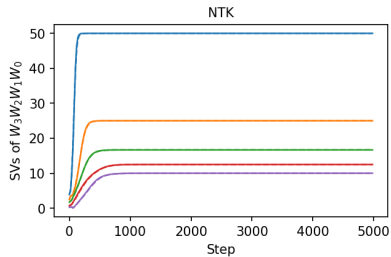
$\gamma < 1$: large initialization, linear regime (NTK)

$\gamma > 1$: small initialization, saddle-to-saddle

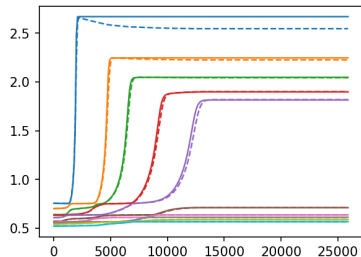
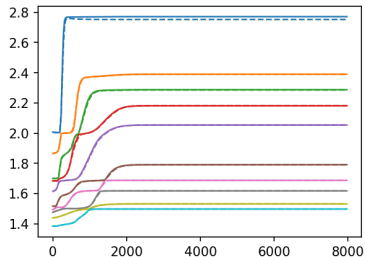
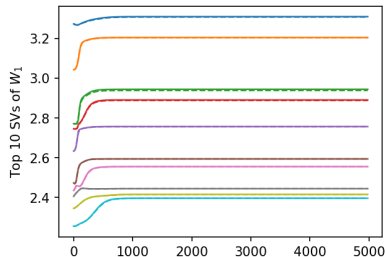
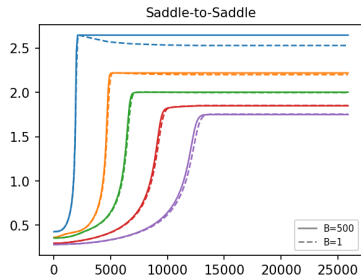
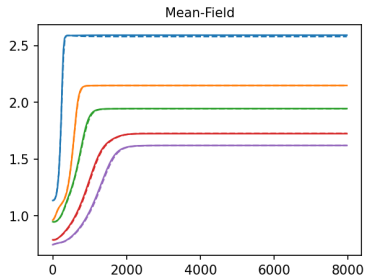
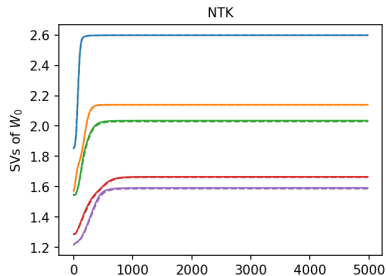
$\gamma = 1$: intermediate regime



Singular values of student matrix



Singular values of individual weight matrices



- Mathematically tractable toy model that exhibits interesting and relevant phenomena
- Some behaviours transfer to nonlinear architectures
- A first model for understanding implicit biases and weight interpretability during training

Geometry

A 2-layer diagonal DLN

Model and loss

Take a 2-layer diagonal DLN of width 2:

$$W_1 = \text{diag}(b_1, b_2), \quad W_2 = \text{diag}(a_1, a_2), \quad \theta = (W_1, W_2).$$

Hence the student matrix is

$$W = W_2 W_1 = \text{diag}(a_1 b_1, a_2 b_2).$$

Let the teacher be

$$M = \text{diag}(s_1, s_2), \quad s_1, s_2 > 0, \quad x \sim \mathcal{N}(0, I).$$

With whitened inputs, the quadratic population loss is

$$\mathcal{L}(\theta) = \frac{1}{2} \|M - W\|_F^2 = \frac{1}{2} (a_1 b_1 - s_1)^2 + \frac{1}{2} (a_2 b_2 - s_2)^2.$$

Gradient equations

For $i = 1, 2$,

$$\partial_{a_i} \mathcal{L} = (a_i b_i - s_i) b_i, \quad \partial_{b_i} \mathcal{L} = (a_i b_i - s_i) a_i.$$

Therefore the critical point equations are

$$(a_i b_i - s_i) b_i = 0, \quad (a_i b_i - s_i) a_i = 0, \quad i = 1, 2.$$

The two coordinates decouple: each singular direction can be analyzed independently.

Coordinate-wise characterization

For each $i = 1, 2$, the critical point equations are

$$(a_i b_i - s_i) b_i = 0, \quad (a_i b_i - s_i) a_i = 0.$$

Since $s_i > 0$, this implies exactly two possibilities:

$$a_i b_i = s_i \quad \text{or} \quad a_i = b_i = 0.$$

Thus each coordinate either fits the teacher singular value or is shut off.

Critical points

All critical points

Parameter conditions	Realized student matrix W	Loss
$a_1 b_1 = s_1, a_2 b_2 = s_2$	$W = \text{diag}(s_1, s_2) = M$	$\mathcal{L} = 0$
$a_1 b_1 = s_1, a_2 = b_2 = 0$	$W = \text{diag}(s_1, 0)$	$\mathcal{L} = \frac{1}{2} s_2^2$
$a_1 = b_1 = 0, a_2 b_2 = s_2$	$W = \text{diag}(0, s_2)$	$\mathcal{L} = \frac{1}{2} s_1^2$
$a_1 = b_1 = 0, a_2 = b_2 = 0$	$W = 0$	$\mathcal{L} = \frac{1}{2}(s_1^2 + s_2^2)$

Critical points correspond to choosing a subset of the teacher singular values.
Depth introduces three new critical points compared with the one-layer DLN loss

$$\mathcal{L}(W) = \|M - \text{diag}(w_1, w_2)\|_F^2.$$

Global minima and saddle point

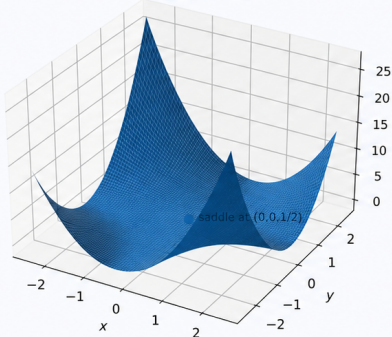
Loss: $\ell_1(x, y) = \frac{1}{2}(xy - 1)^2$

Global minima: $\{(x, y) : xy = 1\}, \ell_1 = 0$

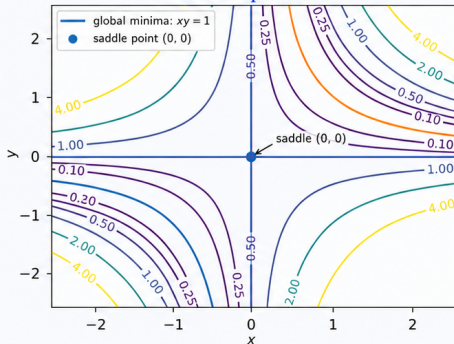
Saddle point: $(0, 0), \ell_1(0, 0) = \frac{1}{2}$

Strict saddle: along $x = y = t$ the loss decreases near 0, while along $x = -y = t$ the loss increases near 0

3D loss surface



Contour plot



Global minima form the hyperbola $xy = 1$, while $(0, 0)$ is a strict saddle point.

Deep learning without poor local minima

Assume that Σ_X and Σ_{YX} are full rank, and that Σ_{YX} has d_L distinct singular values.

No non-global local minima

The loss function of a DLN of depth L is non-convex and non-concave, and every critical point that is not a global minimum is a saddle point.

The training dynamics cannot get stuck in spurious local minima.

Reference: *Kawaguchi, Kenji. "Deep learning without poor local minima." Advances in neural information processing systems 29 (2016)*

Parametrization of critical points

Teacher SVD: $M = U \operatorname{diag}(s_1, \dots, s_{r_m}, 0_{d_L - r_m}) V^\top$.

Saddle points

Let $\theta \in \Omega$ be a critical point of rank r . Then there exists a subset S of left singular vectors of M and a projector $P_S = U_S U_S^\top$ onto their span such that:

$$\mu(\theta) = P_S M = U_S \operatorname{diag}(s_1, \dots, s_r, 0_{d_L - r}) V^\top$$

The saddle points are given by a subset of singular values and (left) singular vectors of the teacher M .

Reference: *Achour, El Mehdi, François Malgouyres, and Sébastien Gerchinovitz. "The loss landscape of deep linear neural networks: a second-order analysis." Journal of Machine Learning Research 25.242 (2024): 1-76.*

Symmetries

Let $G_d := \prod_{\ell=0}^L GL_{d_\ell}$ act on Ω via:

$$(W_1, \dots, W_L) \mapsto (g_1 W_1 g_0^{-1}, g_2 W_2 g_1^{-1}, \dots, g_L W_L g_{L-1}^{-1}),$$

Equivariance of the multiplication map

Since $W_L g_{L-1}^{-1} g_{L-1} W_{L-1} \cdots g_1^{-1} g_1 W_1 = W_L \cdots W_1$, the multiplication map μ is G_d -equivariant and G_h -invariant:

$$\mu(g \cdot \theta) = g_L W g_0^{-1}, \quad \mu(g_h \cdot \theta) = \mu(\theta).$$

A DLN is degenerate: infinitely many DLNs can map to the same student matrix.

It is natural to study solutions up to G_d symmetries. By equivariance of μ , G_d acts on the determinantal variety

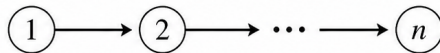
$$\Sigma_d^r := \{\theta \in \Omega \mid \text{rank}(\mu(\theta)) = r\}.$$

Quiver representation theory

Basic notions

- Quiver: directed graph $Q = (I, E, s, t)$
- vertices I , arrows E
- for $e \in E$: $s(e) = \text{source}$, $t(e) = \text{target}$
- loops and cycles allowed
- Representation of Q :
 - assign a vector space V_i to each vertex $i \in I$
 - assign a linear map ϕ_e to each arrow $e \in E$
- Dimension vector: $d = (d_i)_{i \in I}$
- $\text{Rep}_d(Q)$: space of representations of dimension vector d

Example: the linear quiver A_n



Example representation: an interval module

$$0 \xrightarrow{0} \cdots \xrightarrow{0} 0 \xrightarrow{0} k \xrightarrow{\text{id}} \cdots \xrightarrow{\text{id}} k \xrightarrow{0} 0 \xrightarrow{0} \cdots \xrightarrow{0} 0$$

DLN as quiver representations

The multiplication map of a DLN is a map $\mu : \text{Rep}_d(A_{L+1}) \rightarrow \text{Mat}_{d_L \times d_0}$,

with vector spaces $\mathbb{R}^{d_i}$ at the vertices and linear maps W_i on the arrows.

In particular, Σ'_d is a subvariety of $\text{Rep}_d(A_{L+1})$.

Classification of indecomposables of $\text{Rep}_d(A_{L+1})$.

An indecomposable is a representation M_i in $\text{Rep}_d(Q)$ that cannot be written as a non-trivial direct sum; every $M \in \text{Rep}_d(Q)$ decomposes as $M \cong \bigoplus_i M_i^{\oplus m_i}$.

Indecomposables of A_L

$$k^2 \xrightarrow{\phi_1} k^2 \xrightarrow{\phi_2} k \cong k \xrightarrow{id} k \xrightarrow{id} k \oplus k \xrightarrow{id} k \xrightarrow{0} 0.$$

In general, the indecomposable **interval module**

$$M(i, j) = 0 \xrightarrow{0} \cdots \xrightarrow[0]{i} k \xrightarrow{id} \cdots \xrightarrow{id} k \xrightarrow[0]{j+1} 0 \rightarrow \cdots \xrightarrow{0} 0$$

$$0 \leq i < j \leq L + 1.$$

Decomposition via rank patterns

$$M \cong \bigoplus_{r_{ij} \in \mathcal{R}_d} M(i, j)^{\oplus r_{ij}}$$

With $r_{ij} := \text{rank}(W_{i \rightarrow j})$ and $r_{ii} = d_i$ the *rank patterns*; and $\mathcal{R}_d$ the set of rank patterns.

Two representations are isomorphic iff they are in the same $\mathcal{G}_d$ orbit iff they have the same rank pattern.

Stratification of the determinantal variety

Orbits are parameterized by rank patterns

The G_d -orbits in Σ_d^r are indexed by rank patterns:

$$\Sigma_d^r = \bigsqcup_{\rho \in \mathcal{R}_d^r} \mathcal{O}_\rho$$

Two representations are in the same G_d -orbit if and only if they are isomorphic, equivalently if and only if they have the same decomposition into indecomposables.

Thus rank patterns stratify Σ_d^r . Since $\Sigma_d^{r_m} = G_{\text{out}} \cdot \mu^{-1}(M)$, this gives a combinatorial description of the global minima.

Rank patterns are natural observable geometric invariants for DLNs.

Codimension of the global minima

For a teacher matrix M of rank r_m , we have:

$$\begin{aligned}\operatorname{codim}_{\operatorname{Rep}_d}(\mu^{-1}(M)) &= \operatorname{codim}_{\operatorname{Rep}_d}(\Sigma_d^{r_m}) + r_m(d_0 - d_L - r_m) \\ &= 2 \operatorname{RLCT}(\mathcal{L}^{-1})(0).\end{aligned}$$

In general, we only have:

$$\operatorname{RLCT}(\mathcal{L}^{-1})(0) \leq \frac{1}{2} \operatorname{codim}_{\operatorname{Rep}_d}(\mathcal{L}^{-1}(0)).$$

Ref: *Lehalleur, Simon Pepin, and Richárd Rimányi. "Geometry of fibers of the multiplication map of deep linear neural networks." arXiv preprint arXiv:2411.19920 (2024).*

Takeaways

Under full-rank and nondegenerate-data assumptions:

- Critical points of DLNs are either saddles or global minima
- Gauge symmetries mean there is a whole algebraic variety of degenerate minima
- Quiver representation theory gives a combinatorial description of the global minima
- The codimension of the global minima is twice the RLCT

- Kenji Kawaguchi. Deep learning without poor local minima *Advances in neural information processing systems*, 29, 2016.
- El Mehdi Achour, François Malgouyres, and Sébastien Gerchinovitz. The loss landscape of deep linear neural networks: a second-order analysis. *Journal of Machine Learning Research*, 25(242):1–76, 2024.
- Anna Choromanska, Mikael Henaff, Michael Mathieu, Gérard Ben Arous, and Yann LeCun. The loss surfaces of multilayer networks. In *Artificial intelligence and statistics*, pages 192–204. PMLR, 2015.
- Lehalleur, Simon Pepin and Rimányi, Richárd. Geometry of fibers of the multiplication map of deep linear neural networks, arXiv preprint arXiv:2411.19920, 2024.
- Chen, Po and Jiang, Rujun and Wang, Peng. A Complete Loss Landscape Analysis of Regularized Deep Matrix Factorization, arXiv preprint arXiv:2506.20344, 2025.

Dynamics

Two-layer DLN

$$W := W_2 W_1,$$

$$R(W) := \Sigma_{YX} - W_2 W_1.$$

For the population loss

$$\mathcal{L}(W_1, W_2) = \frac{1}{2} \|\Sigma_{YX} - W_2 W_1\|_F^2,$$

gradient flow is

$$\dot{W}_1 = W_2^\top R(W),$$

$$\dot{W}_2 = R(W) W_1^\top.$$

Symmetries induce conserved quantities through gradient flow

Hidden gauge symmetry

For a 2-layer DLN, $f_\theta(x) = W_2 W_1 x$ and $\mathcal{L}(W_2, W_1) = \Phi(W_2 W_1)$.

$$G = \text{GL}_h, \quad g \cdot (W_2, W_1) = (W_2 g^{-1}, g W_1).$$

This action leaves the product $W_2 W_1$, hence the loss, invariant.

$$\underbrace{W_2 g^{-1} \times g W_1}_{g \cdot (W_2, W_1)} \longrightarrow W_2 (g^{-1} g) W_1 = W_2 W_1$$

Lie algebra $\rightarrow$ infinitesimal symmetry

For $A \in \mathfrak{gl}_h$ and $g(t) = e^{tA}$,

$$X_A(W_2, W_1) := \left. \frac{d}{dt} \right|_{t=0} (W_2 e^{-tA}, e^{tA} W_1) = (-W_2 A, A W_1).$$

X_A is the infinitesimal symmetry vector field generated by A on parameter space.

$$d\mathcal{L}_\theta(X_A) = 0 \quad \forall A \in \mathfrak{gl}_h$$

$d\mathcal{L}_\theta$ is the differential of the loss at θ ; it acts on tangent vectors.

$$\implies W_1 (\nabla_{W_1} \mathcal{L})^\top = (\nabla_{W_2} \mathcal{L})^\top W_2.$$

Conserved quantity under gradient flow

$$\dot{W}_i = -\nabla_{W_i} \mathcal{L}, \quad \beta(W_2, W_1) := W_2^\top W_2 - W_1 W_1^\top.$$

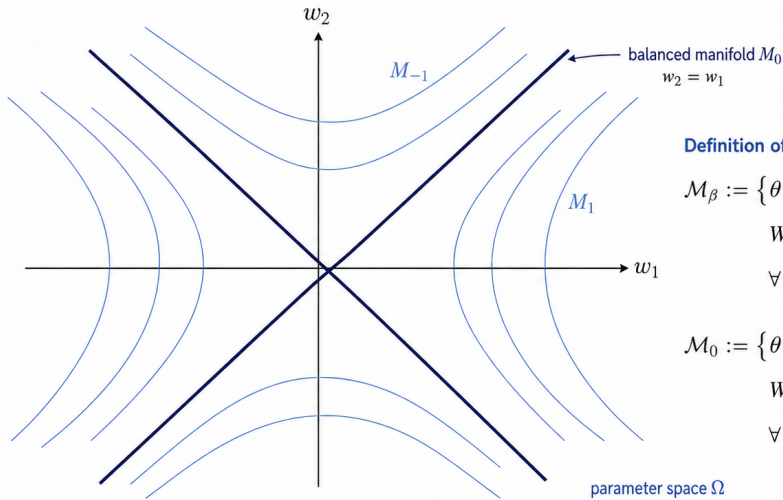
$$\boxed{\frac{d}{dt} \beta(W_2, W_1) = 0}$$

Balanced initialization is preserved:

$$W_2(0)^\top W_2(0) = W_1(0) W_1(0)^\top \implies W_2(t)^\top W_2(t) = W_1(t) W_1(t)^\top.$$

The balanced condition is the Noether-type conserved quantity associated with the hidden GL_h symmetry.

Foliation of parameter space by balance leaves



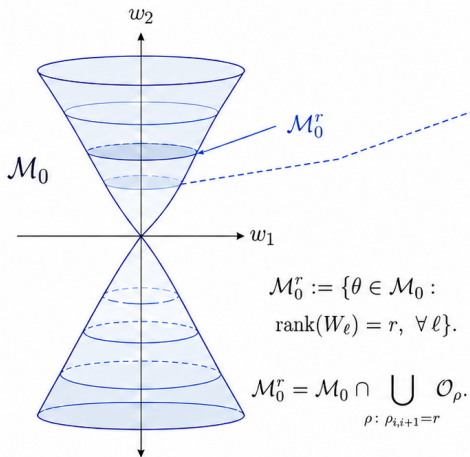
Definition of the balance leaves

$$\mathcal{M}_\beta := \left\{ \theta \in \Omega : \begin{aligned} &W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell, \\ &\forall \ell = 1, \dots, L-1 \end{aligned} \right\}$$

$$\mathcal{M}_0 := \left\{ \theta \in \Omega : \begin{aligned} &W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = 0, \\ &\forall \ell = 1, \dots, L-1 \end{aligned} \right\}$$

► Gradient flow remains on a fixed balance leaf.

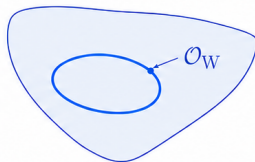
Zooming into the balanced cone



Balanced manifold stratified by fixed internal-layer rank.



$\mathcal{M}_0^r$



$$\mathcal{O}_W := \mu^{-1}(W) \cap \mathcal{M}_0^r \\ = \{\theta \in \mathcal{M}_0^r : \mu(\theta) = W\}.$$

$$\mu|_{\mathcal{M}_0^r} : \mathcal{M}_0^r \rightarrow \mathcal{F}_r, \\ \mathcal{F}_r := \{W \in \mathcal{F} : \text{rank}(W) = r\}.$$

On each stratum, the restricted product map is a submersion onto the rank- r student variety $\mathcal{F}_r$.

$$\text{microstates} = \text{Vol}(\mathcal{O}_W), \\ s(W) = \log \text{Vol}(\mathcal{O}_W).$$

$$\mathcal{F}(W) = \mathcal{L}(W) - \beta^{-1}s(W).$$

Degeneracy counts the number of parameter states in the stratum mapping to the same student.

The manifold of gradient flow

Gradient flow is constrained on the following hyperbolic manifold:

$$\mathcal{M}_\beta := \{\theta \in \Omega \mid W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top = \beta_\ell\}$$

Gradient flow on the balanced variety

On the balanced variety $\mathcal{M}_0$, the self-consistent gradient flow equation in function space is

$$\dot{W} = \sum_{k=1}^L (WW^\top)^{\frac{L-k}{L}} R(W) (W^\top W)^{\frac{k-1}{L}}, \quad R(W) := \Sigma_{YX} - W.$$

For depth 2:

$$\mathcal{M}_0 = \{(W_1, W_2) \mid W_2^\top W_2 = W_1 W_1^\top\}.$$

$$\dot{W} = (WW^\top)^{\frac{1}{2}} R(W) + R(W) (W^\top W)^{\frac{1}{2}}.$$

Balanced parameterization of the orbit

Balanced parameterization

$$W_p = Q_p \Lambda Q_{p-1}^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \quad \lambda_i = \sigma_i(W)^{1/L}.$$

$$W = Q_L \Sigma Q_0^T.$$

Fixing W fixes Q_L, Λ, Q_0 ; the hidden orthogonal matrices $Q_1, \dots, Q_{L-1}$ remain free.

$$\Theta_W = O(d)^{L-1}$$

hidden orthogonal
matrices

Ψ_W



$$O_W$$

Map definition:

$$\Psi_W(Q_1, \dots, Q_{L-1}) = (W_L, \dots, W_1),$$

$$W_p = Q_p \Lambda Q_{p-1}^T.$$



$$O_W \simeq O(d)^{L-1}.$$



The hidden gauge freedom is carried by the orthogonal matrices $Q_1, \dots, Q_{L-1}$.

Orbit volume

Two-layer example ($L=2$)

$$\begin{aligned} W_2 &= U \Lambda Q^\top, & W_1 &= Q \Lambda V^\top. \\ Q(t) &= e^{tA} Q, & A &= x e_{ij}, \quad e_{ij} \in o(d). \\ \delta W_2 &= -x \Lambda e_{ij}, & \delta W_1 &= x e_{ij} \Lambda. \\ \|\delta W\|_F^2 &= x^2 (\|\Lambda e_{ij}\|_F^2 + \|e_{ij} \Lambda\|_F^2). \end{aligned}$$

$$\begin{aligned} \|\delta W\|_F^2 &= x^\top H_{ij} x \\ H_{ij} &= [\sigma_i + \sigma_j], \\ \sqrt{\det H_{ij}} &= \sqrt{\sigma_i + \sigma_j}. \end{aligned}$$



General depth

$$\begin{aligned} \text{dvol}_{O_W} &= \prod_{i < j} \sqrt{\det H_{ij}} \, dx \\ \text{vol}(O_W) &= c_d^{L-1} \prod_{i < j} \sqrt{\det H_{ij}} \\ s(W) &= \log \text{vol}(O_W) \\ \mathcal{F}(W) &= \mathcal{L}(W) - \beta^{-1} s(W) \end{aligned}$$

References

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Saddle-to-saddle regime in DLNs

Assume whitened inputs $\Sigma_X = I$ and $\Sigma_{YX} = USV^\top$. Define the layerwise modes of a 2-layer DLN by

$$w_1^\alpha := W_1 v^\alpha \in \mathbb{R}^{d_1}, \quad w_2^\alpha := W_2^\top u^\alpha \in \mathbb{R}^{d_1},$$

The DLN mode matrix is

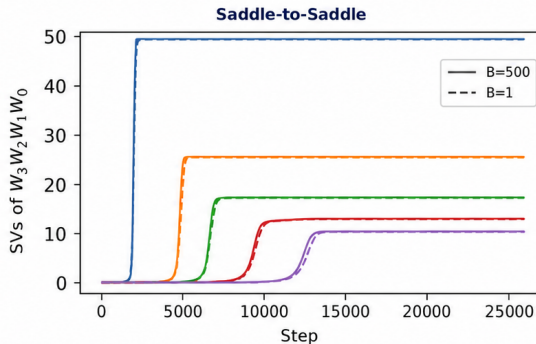
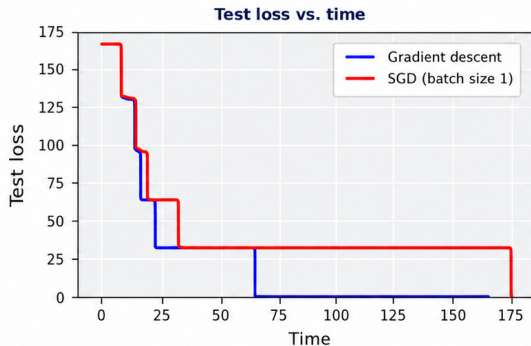
$$w_{\alpha\beta} = u^{\alpha\top} W_2 W_1 v^\beta = (w_2^\alpha)^\top w_1^\beta.$$

Assuming gradient flow on the balanced manifold and using the self-consistent gradient-flow equation:

$$\dot{w}_{\alpha\alpha} = 2(s_\alpha - w_{\alpha\alpha}) w_{\alpha\alpha} - 2 \sum_{\beta \neq \alpha} (w_{\alpha\beta})^2.$$

Rich regime: balanced, aligned weights

$$\dot{w} = 2(s - w)w$$



Plateaus in the loss correspond to learning modes one by one.

$$T = \frac{1}{2s} \ln \left(\frac{w_f (s - w_0)}{w_0 (s - w_f)} \right)$$

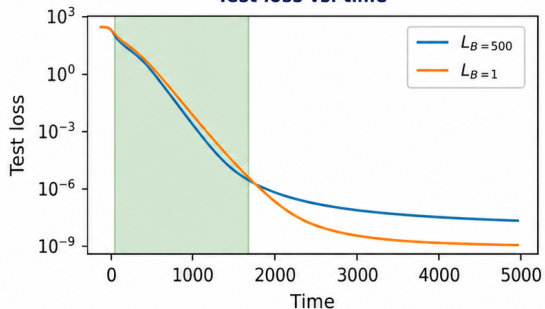
w_0 : initial value

w_f : final value

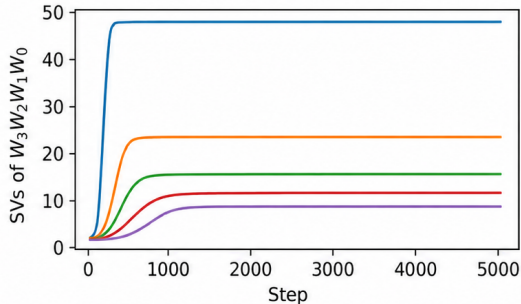
Linear regime

$$\dot{w} = k_0(s - w)$$

Test loss vs. time



NTK



In the linear regime, all modes start growing early and the loss decays smoothly without pronounced plateaus.

$$t \approx -k_0^{-1} \log \left(\frac{s - w_f}{s - w_0} \right)$$

w_0 : initial value

w_f : final value

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- DLNs are mathematically tractable and informative about deep learning
- Useful for understanding implicit biases from depth, scale, overparameterization, hyperparameters, and optimizers
- Symmetries and conserved quantities are central
- Different regimes of training dynamics: rich vs. lazy
- However, they are limited: they cannot learn nonlinear compositional structure

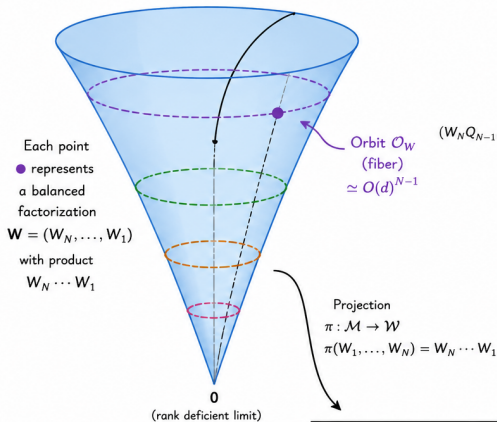
Research directions

- Combinatorial description of all critical points (work in progress)
- Generalize the volume formula to the rank-deficient case
- Volume formula for the transient saddle-to-saddle regime
- Loss landscapes of nonlinear networks (ReLU)
- Geometric invariants for nonlinear networks
- Toy models of hierarchical learning (nonlinearities)

Topics we did not discuss:

- Data distribution
- Nonlinearities (gated DLNs)
- Stochasticity
- Discrete learning rate
- Scale and statistical field theory

Balanced weight manifold $\mathcal{M}$ (a cone)

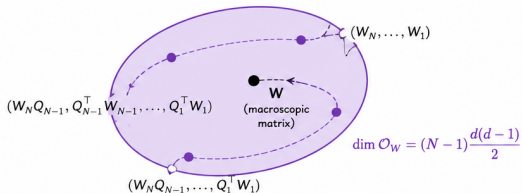


Legend

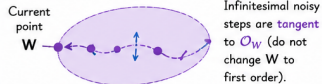
- Orbit $\mathcal{O}_W$ (fiber)
- Balanced manifold $\mathcal{M}$
- Quotient space $\mathcal{W} = \mathcal{M}_d$
- Balanced factorization $\mathbf{W}$

Orbit $\mathcal{O}_W$ over a fixed $\mathbf{W}$

All points in $\mathcal{O}_W$ give the same product $\mathbf{W}$.
They differ by internal orthogonal transformations.



Fluctuations along the orbit (gauge noise)



Effect of fluctuations (entropic drift)

